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Analysing Syntactic Long-term Alignment in Large Language Models

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Abstract

I hereby confirm that the thesis presented here is my own work, with all assistance acknowledged. I assure that the electronic version is identical in content to the printed version of the thesis.

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Abstract

This thesis explores the effects of long-term syntactic alignment in Large Language Models (LLMs). An approach by Reitter (2008) was adapted to analyze alignment effects between speakers. The analysis replicates previous findings from Reitter & Moore and was validated against baseline data. Additionally, new datasets created with OpenAI's GPT-4o provide a basis for analyzing conversational texts generated by LLMs, addressing a gap in existing research for evaluating LLM-generated texts.

The findings reveal that syntactic alignment occurs in LLMs over extended periods. A second analysis further explores how the process of alignment evolves throughout a conversation, showing that LLMs consistently adjust their syntax, with the largest changes occurring early on.

These results indicate that LLMs are not only influenced by the semantics of instructions, but also by the syntax present in their context. LLM responses are shaped not only by the vector representations of tokens in their prompts, but also by the linear order in which those tokens appear.

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Chapter 1

Introduction

Large language models are commonly evaluated by asking whether they solve tasks, follow instructions, or provide useful answers. This thesis takes a complementary perspective. It asks whether the form of a model's language is shaped by the form of the language that precedes it. In particular, I study syntactic long-term alignment: the tendency of a system to reuse, avoid, or gradually drift toward syntactic patterns that appeared earlier in a dialogue.

The phenomenon is familiar from human conversation. Speakers often converge on similar lexical choices, comparable referring expressions, and parallel syntactic frames. Such alignment is rarely perfect copying. It is better understood as a pressure on the distribution of choices available to a speaker. If one participant repeatedly uses a relative clause or a passive construction, the other participant may become slightly more likely to use the same kind of construction later.

For language models, this question is both practical and theoretical. It is practical because many applications place models in long conversations, tutoring sessions, editing workflows, or interactive coding environments. In those settings, syntactic drift can change the perceived style of the assistant. It is theoretical because alignment effects give us a way to ask how much of a model's behavior is driven by abstract task instructions and how much is driven by the surface form of the prompt. This distinction echoes broader debates about form, meaning, and understanding in neural language systems (Bender and Koller 2020).

The rest of the thesis is organized as follows. Chapter 2 reviews syntactic alignment in human dialogue and summarizes related work on formal syntax and model evaluation. Chapter 3 introduces the datasets and the annotation pipeline. Chapter 4 presents the main alignment experiments. Chapter 5 discusses qualitative patterns, limitations, and implications for prompt design.

Chapter 2

Background

The study of alignment begins with the observation that dialogue is coordinated at many levels. Participants coordinate topics, turn boundaries, referring expressions, lexical choices, and grammatical forms. This coordination reduces effort: once a construction has been used, it becomes easier for both speakers to reuse it. Psycholinguistic work often explains this with priming, shared representations, or interactive alignment mechanisms.

In computational settings, alignment is harder to define. A language model has no stable communicative intention in the human sense. It samples or selects continuations from a distribution conditioned on context. Nevertheless, the output distribution can reflect structural properties of earlier text. If a prompt contains many subordinate clauses, later responses may become more likely to contain subordinate clauses as well.

The difficulty is that surface similarity can be misleading. Two responses may both contain long sentences without sharing a syntactic construction. Conversely, two responses can differ lexically while sharing an abstract dependency pattern. For this reason, the experiments in this thesis use parsed representations rather than raw string overlap.

2.1 Related Work

Syntactic priming has been studied extensively in human production. Classic experiments show that speakers are more likely to reuse a syntactic frame after hearing or producing it. The effect has been observed in dialogue, picture description, and sentence completion tasks. It is usually small but reliable, and it interacts with lexical repetition, semantic similarity, and discourse context.

Work on neural language models has focused more often on semantic consistency, instruction-following, and factuality. However, several strands of research are relevant here. Prompt sensitivity studies show that small changes in wording can change model behavior. Style transfer and persona consistency studies show that models can preserve surface features across a generation. Long-context evaluations show that models can retrieve and reuse details from earlier passages.

The analysis also draws on formal accounts of grammar. Combinatory Categorial Grammar is one example of a framework in which lexical entries and combinatory rules jointly determine the space of possible derivations (Kuhlmann et al. 2015). The experiments below do not implement a CCG parser, but the general point is important: syntactic form is not merely a string-level property. It involves structured dependencies that must be approximated carefully when measured automatically.

Phenomenon	Typical signal	Timescale
Lexical reuse	Repeated words or phrases	Local to medium
Syntactic priming	Repeated abstract structures	Local
Style matching	Register, sentence length, punctuation	Medium
Long-term alignment	Persistent structural bias	Medium to long

Table 2.1: Summary of alignment-related phenomena.

Chapter 3

Data and Methods

The experiments use synthetic conversations designed to isolate syntactic form from semantic content. Each conversation consists of a neutral task description, a context paragraph, and a probe question. The context paragraph appears in one of two variants. The variants express the same content but differ in the target construction. The probe question then asks for a response that is semantically compatible with both variants.

This design makes it possible to compare responses across matched pairs. If the model sees the passive-heavy variant, does it produce more passive clauses in the answer than when it sees the active-heavy variant? If it sees a context with many relative clauses, does it produce more relative clauses in a later explanation?

3.1 Prompt Construction

Prompts were written in pairs. Each pair has a high-alignment condition and a low-alignment condition. The high-alignment condition contains repeated instances of the target construction. The low-alignment condition uses alternative constructions with the same propositional content. The probe is identical across the pair.

For example, a passive condition might describe several observations as “the file was checked”, “the result was recorded”, and “the summary was revised”. The matched active condition would use “the assistant checked the file”, “the script recorded the result”, and “the student revised the summary”. The final probe asks for an explanation of the procedure without mentioning voice.

The prompts were also balanced for length. Exact equality is impossible because constructions have different word counts, but the average difference between paired contexts was kept below five percent. Pairs with larger differences were revised or removed.

3.2 Target Constructions

The analysis focuses on four construction families: active-passive alternations, relative clauses, complement clauses, and prepositional phrase sequences. These constructions were chosen because they are common enough to appear naturally in generated text but structured enough to be detected by a parser.

Construction	Detection cue	Example
Passive voice	Auxiliary plus passive participle	was revised
Relative clause	Relative marker with clausal dependent	that failed
Complement clause	Clausal complement relation	said that…
PP sequence	Two or more adjacent prepositional phrases	in the file on Monday

Table 3.1: Target constructions used in the experiments.



Figure 3.1: Schematic contrast between matched prompt variants.



Figure 3.2: Distribution of response lengths after filtering.

The parser labels are not perfect proxies for linguistic categories. For this reason, each construction family is treated as a measurable approximation. The interpretation of results is therefore conservative: a significant effect indicates a change in parser-detected structure, not necessarily a complete linguistic theory of the model's syntax.

3.3 Generation Protocol

Each prompt was sent to the model with temperature set to a moderate value. Very low temperature makes outputs too deterministic and can hide distributional effects. Very high temperature adds noise and makes parsing less reliable. The chosen setting balances variation with coherence.

For each prompt pair, multiple responses were generated. Responses were discarded if they failed to answer the probe, contained fewer than a minimum number of tokens, or copied a large fragment of the context. The remaining responses were parsed and converted into construction counts.

The primary outcome is the difference in normalized construction rate between the aligned and control conditions. Secondary outcomes include response length, parse failure rate, and qualitative error categories. As in many model-evaluation settings, the analysis treats performance estimates as noisy signals rather than final judgments (Yao and Koller 2023).

3.4 Annotation Pipeline

After generation, each response passes through an annotation pipeline that turns text into counts of syntactic events. The pipeline has four stages. First, responses are normalized by removing system artifacts and repeated whitespace. Second, each sentence is parsed with a dependency parser. Third, construction-specific detectors collect candidate spans. Finally, aggregate counts are written to a table keyed by prompt pair, condition, construction family, and model run.

The pipeline is deliberately conservative. It is better to miss a few borderline cases than to inflate the alignment signal with false positives. For example, a passive detector should not count every past participle as passive. It should require evidence for an auxiliary and a dependency relation compatible with passive voice. Similarly, a relative-clause detector should distinguish finite relatives from ordinary complement clauses where possible.

3.4.1 Normalization

Normalization is small but important. Model responses often contain leading discourse markers such as “Sure” or “Here is a concise explanation”. These phrases are not removed, because they are part of the generated response. However, boilerplate that repeats the prompt or explicitly describes the task is filtered. The goal is to preserve natural syntax while excluding obvious failures of the generation protocol.

Responses are also segmented into sentences. Sentence segmentation is treated as a diagnostic rather than a ground truth operation. If a response contains an unusually high number of fragments, the parser may still produce an analysis, but the response is flagged for later inspection.

3.4.2 Detector Rules

Each construction family is represented by a small set of detector rules. These rules combine parser labels with lexical and morphological cues. A passive construction, for instance, is counted when the parser marks a passive auxiliary or passive subject relation. A relative clause is counted when a clausal modifier is introduced by a relative marker or has the dependency shape expected for a relative clause.

Passive voice. Passive voice is the most reliable construction in the sample because the parser usually marks the relevant dependencies explicitly. The detector counts canonical passives such as “the summary was revised” and ignores adjectival participles unless there is an auxiliary or subject configuration suggesting an eventive reading.

Ambiguous participles. Ambiguous participles are handled cautiously. A phrase such as “the revised summary” is not counted as passive, because it functions as an attributive modifier. A phrase such as “the summary was revised by the assistant” is counted, because the auxiliary and optional by-phrase make the passive analysis clear. This level-4 heading is included to exercise paragraph-style heading behavior in the template.

Relative clauses. Relative clauses are harder to detect because generated text often omits explicit relative markers. The detector therefore distinguishes explicit relatives, such as “the file that failed”, from reduced relatives, such as “the file stored on the server”. Only explicit relatives are used in the main analysis. Reduced relatives are recorded separately and inspected qualitatively.

Example	Decision	Reason
the summary was revised	Count	Auxiliary plus passive participle
the revised summary	Skip	Attributive participle
the file that failed	Count	Explicit relative marker
the file stored locally	Inspect	Reduced relative or modifier

Table 3.2: Detector decisions for ambiguous examples.



Figure 3.3: Manual validation workflow for parser-derived counts.

The separation keeps the primary metric easier to interpret. It also avoids conflating relative clauses with ordinary postnominal modifiers. This is especially useful in technical prose, where postnominal participial phrases are frequent.

3.4.3 Reliability Checks

The annotation pipeline includes several reliability checks. A random sample of responses is inspected manually, and the manual notes are compared with detector outputs. The goal is not to produce a full gold standard, but to identify systematic detector failures before interpreting the aggregate results.

The most common errors involve sentence boundaries and reduced clauses. Sentence-boundary errors can make a construction appear to span more material than it actually does. Reduced clauses can be hard to distinguish from adjectival modifiers. These cases are flagged rather than silently folded into the main counts.

3.5 Dataset Summary

The final sample contains prompt pairs distributed across four construction families. Each pair contributes one aligned condition and one control condition. Multiple responses are generated for each condition, which makes it possible to estimate variation across runs as well as across prompt items.

The dataset is intentionally small enough to inspect. Large automatically generated datasets can hide annotation failures, while very small datasets make effect estimates unstable. The sample size used here is meant to balance those concerns for a thesis-scale experiment.

Construction	Pairs	Responses	Filtered
Passive voice	40	480	22
Relative clauses	40	480	31
Complement clauses	32	384	19
PP sequences	32	384	27

Table 3.3: Mock dataset size after filtering.

Chapter 4

Results

The results show consistent but construction-dependent alignment. Passive voice and relative clauses produce the clearest effects. Complement clauses show a smaller effect, and prepositional phrase sequences are noisy. These differences are consistent with the intuition that some constructions are easier for the model to track and reproduce than others.

4.1 Main Alignment Effects

The largest effect appears for passive voice. In passive-rich contexts, the model produces more passive clauses in the probe response than in matched active contexts. The difference is not large in absolute terms, but it is stable across prompt families.

Relative clauses show a similar pattern. Contexts with repeated relative clauses increase the rate of relative clauses in later responses. Qualitative inspection suggests that the model often uses relative clauses to maintain the explanatory style of the context.

The estimates in Table 4.1 are intentionally modest. The important point is not that the model becomes a mirror of the prompt. Rather, the distribution shifts. In an interactive setting, such shifts can accumulate across turns and contribute to the perceived style of the system.

4.2 Persistence Across Turns

Alignment decays with distance, but it does not vanish immediately. When one filler turn is inserted between the context and the probe, the passive effect remains visible. With three filler turns, the effect becomes smaller and less reliable. Relative clauses show a similar but slightly faster decay.

This pattern suggests that syntactic information is retained as part of the recent dialogue state. It is not merely a token-local continuation effect. At the same time, the decay indicates that syntax competes with newer content and instructions.

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Figure 4.1: Schematic decay of alignment over intervening turns.

Construction	Effect	95% CI	Interpretation
Passive voice	+3.8 pp	[+2.1, +5.4]	Clear
Relative clauses	+2.9 pp	[+1.0, +4.6]	Clear
Complement clauses	+1.2 pp	[−0.1, +2.5]	Weak
PP sequences	+0.6 pp	[−1.4, +2.2]	Unclear

Table 4.1: Estimated alignment effects by construction family.

maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius.

Chapter 5

Conclusion

This thesis has examined syntactic long-term alignment in large language models. The results show that prior syntactic context can influence later model output, especially for passive voice and relative clauses. The effect is moderate, construction-dependent, and sensitive to the distance between context and probe.

The broader conclusion is that prompt form matters. Language models respond not only to the meaning of a prompt but also to its linguistic shape. For evaluation, this means that benchmark prompts should be treated as stylistic artifacts as well as task descriptions. For design, it means that examples can guide models in subtle ways that explicit instructions do not capture.

Future work should test naturally occurring dialogues, additional languages, and more precise syntactic annotations. It should also ask when alignment improves interaction and when it amplifies undesirable style. Understanding this balance will help us build systems that adapt without blindly inheriting every pattern in their context.

Bibliography

Emily M. Bender and Alexander Koller (2020). Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.

Marco Kuhlmann, Alexander Koller, and Giorgio Satta (2015). Lexicalization and Generative Power in CCG. In *Computational Linguistics* 41.2, pages 187–219.

Yuekun Yao and Alexander Koller (2023). Predicting generalization performance with correctness discriminators. ArXiv paper 2311.09422. ArXiv: 2311.09422 [cs.CL].